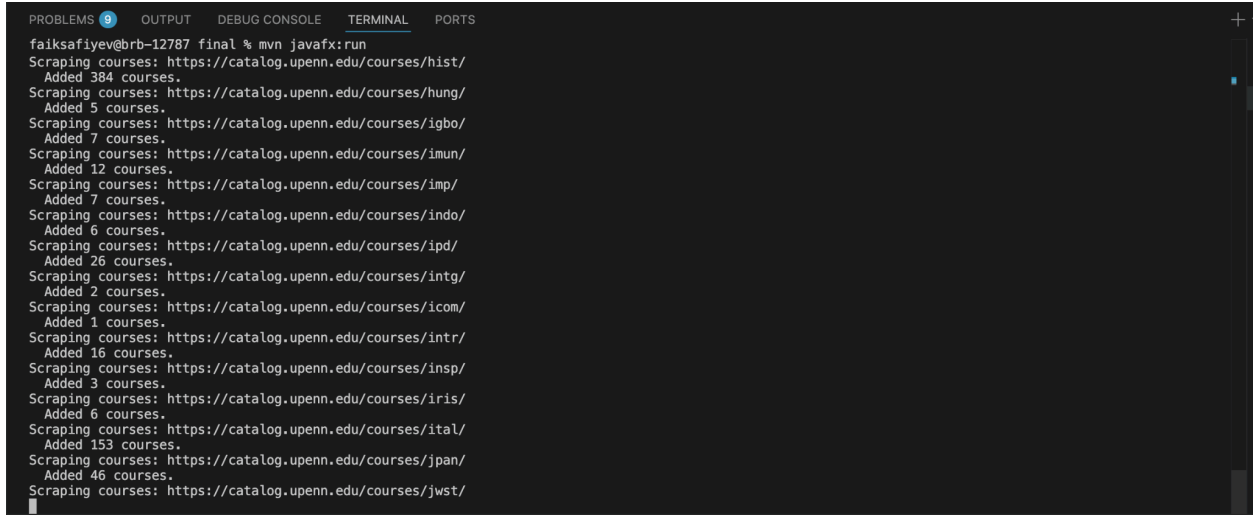


Penn Course Planner User Manual

This document explains how to use our course-scheduling program step-by-step. The application generates a personalized academic schedule based on your preferences, interests, major, minor(s), and course constraints.

Step 1: Launch the Program

Run the program and wait while it scrapes Penn course data in real time. Depending on your internet connection and system performance, this process may take a moment.



```
PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS
faiksafiyev@brb-12787 final % mvn javafx:run
Scraping courses: https://catalog.upenn.edu/courses/hist/
  Added 384 courses.
Scraping courses: https://catalog.upenn.edu/courses/hung/
  Added 5 courses.
Scraping courses: https://catalog.upenn.edu/courses/igbo/
  Added 7 courses.
Scraping courses: https://catalog.upenn.edu/courses/imun/
  Added 12 courses.
Scraping courses: https://catalog.upenn.edu/courses/imp/
  Added 7 courses.
Scraping courses: https://catalog.upenn.edu/courses/indo/
  Added 6 courses.
Scraping courses: https://catalog.upenn.edu/courses/ipd/
  Added 26 courses.
Scraping courses: https://catalog.upenn.edu/courses/intg/
  Added 2 courses.
Scraping courses: https://catalog.upenn.edu/courses/icom/
  Added 1 courses.
Scraping courses: https://catalog.upenn.edu/courses/intr/
  Added 16 courses.
Scraping courses: https://catalog.upenn.edu/courses/insp/
  Added 3 courses.
Scraping courses: https://catalog.upenn.edu/courses/iris/
  Added 6 courses.
Scraping courses: https://catalog.upenn.edu/courses/ita/
  Added 153 courses.
Scraping courses: https://catalog.upenn.edu/courses/jpan/
  Added 46 courses.
Scraping courses: https://catalog.upenn.edu/courses/jwst/
```

Screenshot 1 - Step 1: Launch the Program

Step 2: Enter Your Major

Once the user interface opens, enter your primary major into the typing line of the chatbot.

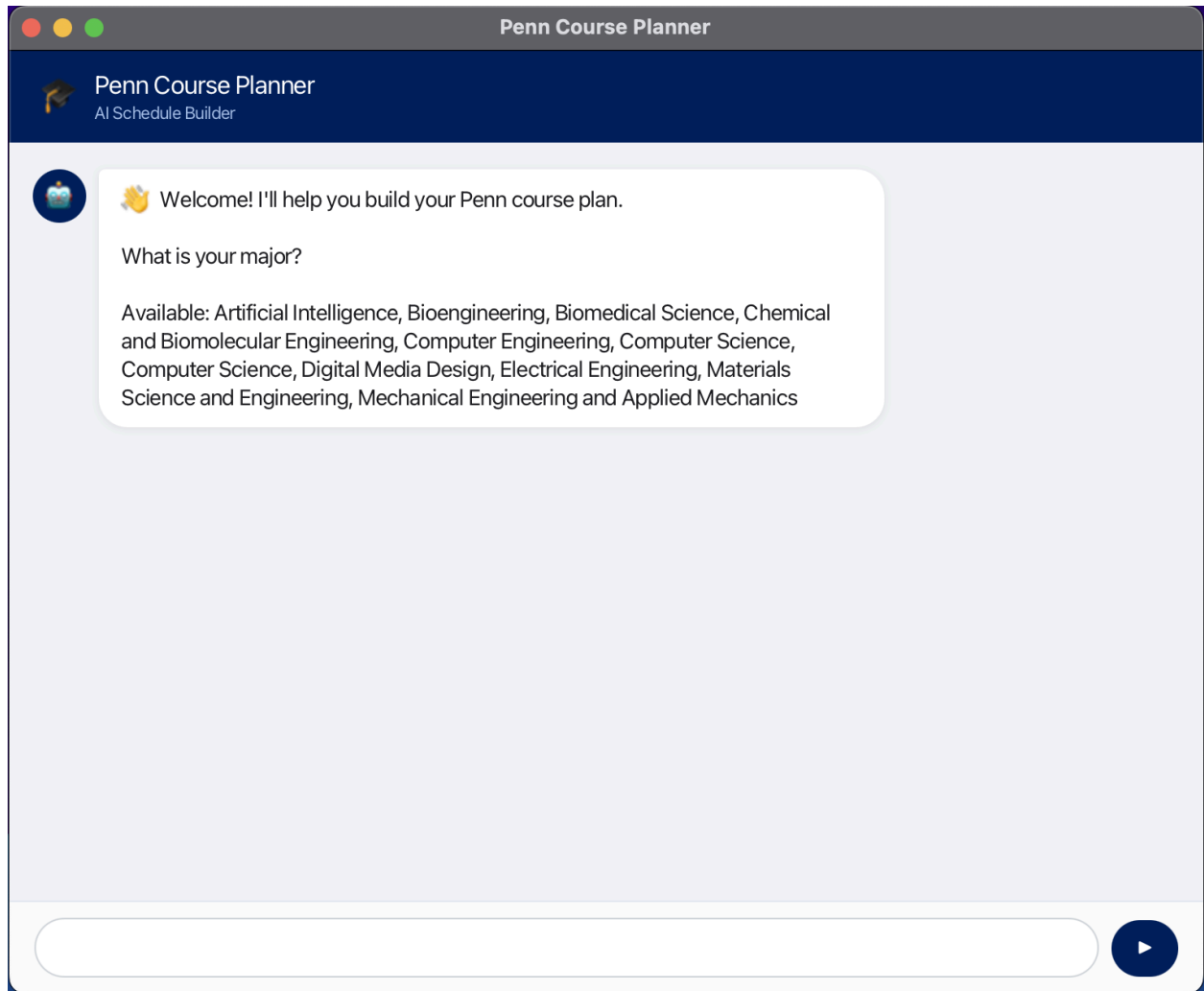
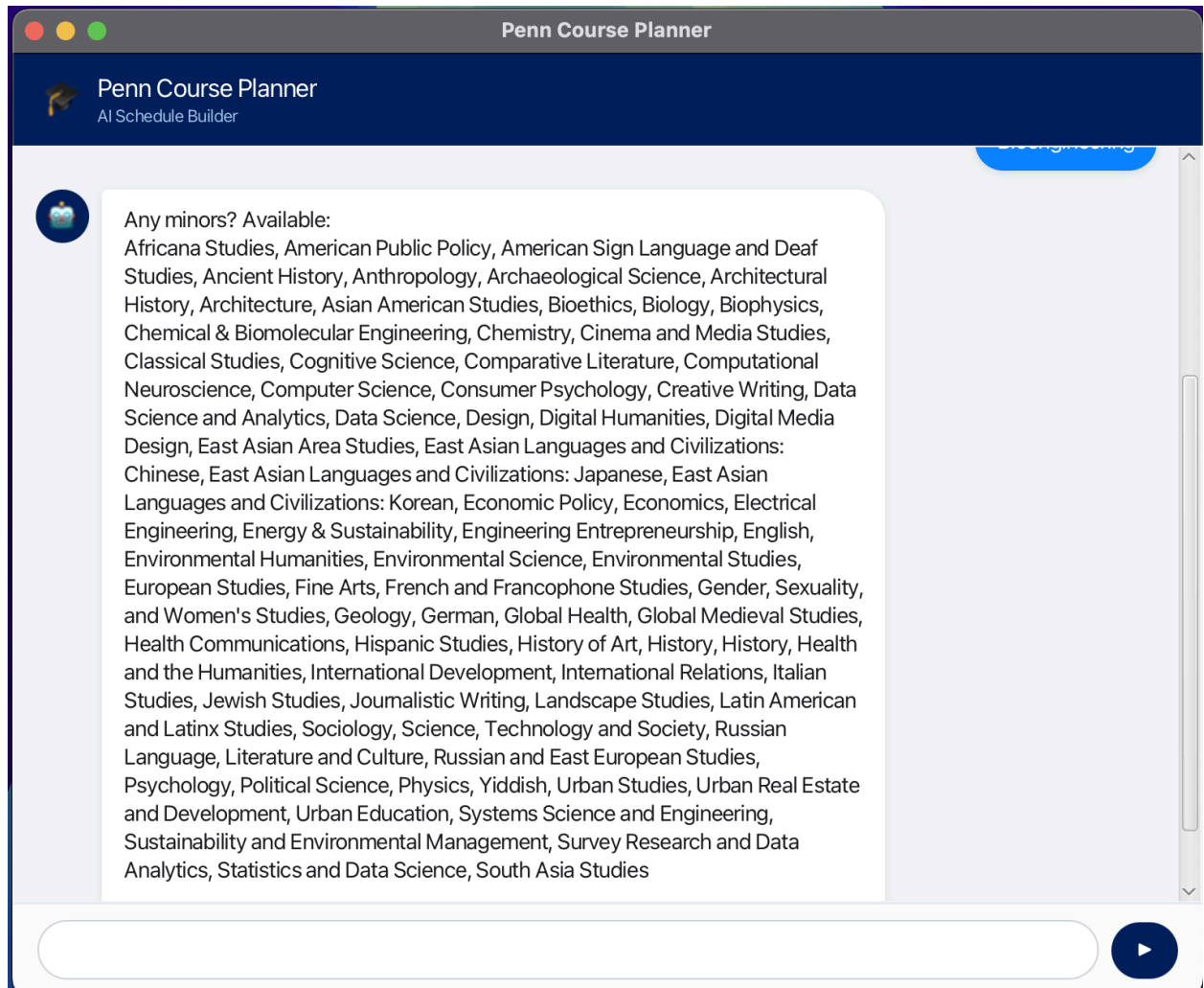


Figure 2 - Step 2: Enter Your Major

Step 3: Enter Your Minor(s)

Input one or more minors you would like to pursue. The system will use this information when generating your potential schedule.



The screenshot shows a web application titled "Penn Course Planner" with a subtitle "AI Schedule Builder". The interface features a dark blue header with a graduation cap icon. Below the header, a light blue box contains a list of minors. To the left of the list is a small icon of a graduation cap. At the bottom of the interface is a white input field and a blue play button icon.

Penn Course Planner
AI Schedule Builder

Any minors? Available:

Africana Studies, American Public Policy, American Sign Language and Deaf Studies, Ancient History, Anthropology, Archaeological Science, Architectural History, Architecture, Asian American Studies, Bioethics, Biology, Biophysics, Chemical & Biomolecular Engineering, Chemistry, Cinema and Media Studies, Classical Studies, Cognitive Science, Comparative Literature, Computational Neuroscience, Computer Science, Consumer Psychology, Creative Writing, Data Science and Analytics, Data Science, Design, Digital Humanities, Digital Media Design, East Asian Area Studies, East Asian Languages and Civilizations: Chinese, East Asian Languages and Civilizations: Japanese, East Asian Languages and Civilizations: Korean, Economic Policy, Economics, Electrical Engineering, Energy & Sustainability, Engineering Entrepreneurship, English, Environmental Humanities, Environmental Science, Environmental Studies, European Studies, Fine Arts, French and Francophone Studies, Gender, Sexuality, and Women's Studies, Geology, German, Global Health, Global Medieval Studies, Health Communications, Hispanic Studies, History of Art, History, History, Health and the Humanities, International Development, International Relations, Italian Studies, Jewish Studies, Journalistic Writing, Landscape Studies, Latin American and Latinx Studies, Sociology, Science, Technology and Society, Russian Language, Literature and Culture, Russian and East European Studies, Psychology, Political Science, Physics, Yiddish, Urban Studies, Urban Real Estate and Development, Urban Education, Systems Science and Engineering, Sustainability and Environmental Management, Survey Research and Data Analytics, Statistics and Data Science, South Asia Studies

Figure 3 - Step 3: Enter Your Minor(s)

Step 4: Select Desired Electives

Enter any elective courses or academic areas you are interested in taking.

The screenshot shows a web application titled "Penn Course Planner" with a subtitle "AI Schedule Builder". The main content area is a light blue box containing a long list of course names and academic areas, such as "Science and Analytics", "Data Science", "Design", "Digital Humanities", "Digital Media Design", "East Asian Area Studies", "East Asian Languages and Civilizations: Chinese", "East Asian Languages and Civilizations: Japanese", "East Asian Languages and Civilizations: Korean", "Economic Policy", "Economics", "Electrical Engineering", "Energy & Sustainability", "Engineering Entrepreneurship", "English", "Environmental Humanities", "Environmental Science", "Environmental Studies", "European Studies", "Fine Arts", "French and Francophone Studies", "Gender, Sexuality, and Women's Studies", "Geology", "German", "Global Health", "Global Medieval Studies", "Health Communications", "Hispanic Studies", "History of Art", "History", "History, Health and the Humanities", "International Development", "International Relations", "Italian Studies", "Jewish Studies", "Journalistic Writing", "Landscape Studies", "Latin American and Latinx Studies", "Sociology", "Science, Technology and Society", "Russian Language, Literature and Culture", "Russian and East European Studies", "Psychology", "Political Science", "Physics", "Yiddish", "Urban Studies", "Urban Real Estate and Development", "Urban Education", "Systems Science and Engineering", "Sustainability and Environmental Management", "Survey Research and Data Analytics", "Statistics and Data Science", and "South Asia Studies". Below the list is a text input field with the placeholder "Type one at a time, or done / none to skip." To the right of the input field is a blue button labeled "done". At the bottom left of the main content area is a small icon of a graduation cap. Below the main content area is a light blue box containing a text input field with the placeholder "Any specific courses you want included? Enter a course ID (e.g. CIS 5450), or done to skip." To the right of the input field is a blue button with a white play icon.

Figure 4 - Step 4: Select Desired Electives

Step 5: Exclude Courses

Specify any courses you do not want included in your schedule. This helps the program avoid unwanted subjects.

Penn Course Planner

Penn Course Planner
AI Schedule Builder

European Studies, Fine Arts, French and Francophone Studies, Gender, Sexuality, and Women's Studies, Geology, German, Global Health, Global Medieval Studies, Health Communications, Hispanic Studies, History of Art, History, History, Health and the Humanities, International Development, International Relations, Italian Studies, Jewish Studies, Journalistic Writing, Landscape Studies, Latin American and Latinx Studies, Sociology, Science, Technology and Society, Russian Language, Literature and Culture, Russian and East European Studies, Psychology, Political Science, Physics, Yiddish, Urban Studies, Urban Real Estate and Development, Urban Education, Systems Science and Engineering, Sustainability and Environmental Management, Survey Research and Data Analytics, Statistics and Data Science, South Asia Studies

Type one at a time, or done / none to skip.

done

Any specific courses you want included? Enter a course ID (e.g. CIS 5450), or done to skip.

done

Any courses you want to exclude? Enter a course ID, or done / none to skip.

done

▶

Figure 5 - Step 5: Exclude Courses

Step 6: Enter Your Interests

Provide your academic or personal interests so the program can recommend more relevant courses.

The screenshot shows a web application titled "Penn Course Planner" with a subtitle "AI Schedule Builder". The interface is a chat-style form with a dark blue header. The main content area is light blue and contains four input prompts, each with a blue "done" button to its right. The first prompt lists various fields of study: "Psychology, Political Science, Physics, Hebrew, Urban Studies, Urban Real Estate and Development, Urban Education, Systems Science and Engineering, Sustainability and Environmental Management, Survey Research and Data Analytics, Statistics and Data Science, South Asia Studies". The second prompt asks for specific course IDs. The third prompt asks for courses to be excluded. The fourth prompt asks for academic interests. At the bottom, there is a text input field containing "databases, machine learning, quantum computers" and a blue play button icon.

Penn Course Planner
AI Schedule Builder

Psychology, Political Science, Physics, Hebrew, Urban Studies, Urban Real Estate and Development, Urban Education, Systems Science and Engineering, Sustainability and Environmental Management, Survey Research and Data Analytics, Statistics and Data Science, South Asia Studies

Type one at a time, or done / none to skip.

done

Any specific courses you want included? Enter a course ID (e.g. CIS 5450), or done to skip.

done

Any courses you want to exclude? Enter a course ID, or done / none to skip.

done

What are your academic interests?

Enter one keyword at a time (e.g. machine learning), or done to skip.

databases, machine learning, quantum computers

Figure 6 - Step 6: Enter Your Interests

Step 7: Set Maximum CU per Semester

Choose the maximum number of Course Units (CU) you would like to take each semester.

The screenshot shows a web application titled "Penn Course Planner" with the subtitle "AI Schedule Builder". The interface is a chat-based form with the following elements:

- Header:** "Penn Course Planner" and "AI Schedule Builder" with a graduation cap icon.
- Question 1:** "Any courses you want to exclude? Enter a course ID, or done / none to skip." with a "done" button.
- Question 2:** "What are your academic interests? Enter one keyword at a time (e.g. machine learning), or done to skip." with a "done" button.
- Input:** A blue button containing the text "databases, machine learning, quantum computers".
- Feedback:** A message: "✓ Added 'databases, machine learning, quantum computers'. Add another, or type done." with a "done" button.
- Question 3:** "Max CU units per semester? (e.g. 4.0 or 5.0)" with a blue button containing the number "6".
- Footer:** A large white input field and a dark blue circular button with a white play icon.

Figure 7 - Step 7: Set Maximum CU per Semester

Step 8: Enter Your Starting Year

Select the academic year in which you plan to begin your studies.

The screenshot shows a web application titled "Penn Course Planner" with the subtitle "AI Schedule Builder". The interface is a chat-based form where a user is entering their starting year. At the top, a blue button contains the text "databases, machine learning, quantum computers". Below this, a confirmation message from the AI assistant states: "✓ Added 'databases, machine learning, quantum computers'. Add another, or type done." To the right of this message is a blue button labeled "done". The next question from the AI is "Max CU units per semester? (e.g. 4.0 or 5.0)", with a blue button labeled "6" to its right. The following question is "Are you starting in Fall or Spring?", with a blue button labeled "Fall" to its right. The final question is "What year are you starting? (e.g. 2024)", with a blue button labeled "2027" to its right. At the bottom of the chat area is a text input field with the placeholder "Type your response.." and a blue play button icon to its right. The entire interface is set against a light gray background with a dark blue header.

Figure 8 - Step 8: Enter Your Starting Year

Step 9: Generate and Review Your Schedule

After all fields are completed, allow the program to generate your personalized schedule. Review the results and make adjustments if needed.

Tip: For the best experience, double-check all entered information before generating the schedule. The more accurately you provide your preferences, the better the generated academic

plan will match your goals.

The screenshot displays the 'Penn Course Planner' AI Schedule Builder interface. The main window shows a 'PENN UNDERGRADUATE COURSE PLAN' with a dark background and white text. The plan is organized by semester and credit units (CU).

Fall 2027 [6.0 CU]

- BE 1000 - Introduction to Bioengineering (0.5 CU)
- ENGR 1050 - Introduction to Scientific Computing (1.0 CU)
- MATH 1400 - Calculus, Part I (1.0 CU)
- CHEM 1101 - General Chemistry Laboratory I (1.0 CU)
- CHEM 1102 - General Chemistry Laboratory II (1.0 CU)
- BIOL 1121 - Introduction to Biology - The Molecular Biology of Life (1.0 CU)
- OIDD 3150 - Databases for Analytics (0.5 CU)

Spring 2028 [6.0 CU]

- MATH 1410 - Calculus, Part II (1.0 CU)
- ENM 3750 - Biological Data Science I - Fundamentals of Biostatistics (1.0 CU)
- CRIM 4012 - Machine Learning for Social Science (1.0 CU)
- BE 4000 - Preceptorship in Clinical Bioengineering (1.0 CU)
- PSCI 4811 - Machine Learning in Political Science, Sociology, and Economics (1.0 CU)
- VLST 2090 - Visual Culture through the Computer's Eye (1.0 CU)

Fall 2028 [6.0 CU]

- ENM 2400 - Differential Equations and Linear Algebra (1.0 CU)
- BIOL 1123 - Introductory Molecular Biology Laboratory (1.0 CU)
- BIOL 3310 - Principles of Human Physiology (1.0 CU)
- BIOE 4010 - Introduction to Bioethics (1.0 CU)
- BE 4260 - Immunology for Bioengineers (1.0 CU)
- PHIL 1440 - Machine Fairness: Algorithms and the Theory of Justice (1.0 CU)

Spring 2029 [6.0 CU]

Below the course plan, there is a confirmation message: 'All done! Scroll up to review your plan.' and a text input field with the placeholder 'Type your response..'. A play button icon is visible in the bottom right corner.

Figure 9 - Step 9: Generate and Review Your Schedule